Python Programming for Efficient Breathomics Image Processing and Analysis

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***Abstract*— Breathomics has emerged as a promising frontier in non-invasive medical diagnostics, focusing on the analysis of biomarkers in exhaled breath. This paper presents a hybrid approach utilizing both traditional machine learning (ML) and deep learning (DL) models to classify human breathing patterns—normal, shallow, and deep—through image-based analysis of breath condensation on cold surfaces. The dataset comprised grayscale images labeled based on visual density and spread characteristics of condensation. Traditional ML models, including Support Vector Machines, Random Forest, and Logistic Regression, were trained using statistical features such as mean intensity, texture variation, and condensation area. In parallel, a series of deep learning models—including a custom Convolutional Neural Network (CNN) and state-of-the-art transfer learning architectures like MobileNetV2, InceptionV3, ResNet50, and EfficientNetB0—were trained on raw image data. Evaluation was performed in multiple testing rounds using held-out and out-of-distribution data subsets to measure model robustness and generalizability. Experimental results indicate that DL models significantly outperform traditional classifiers, with InceptionV3 achieving the highest classification accuracy. This work highlights the potential of image-based breathomics for scalable, contactless respiratory monitoring applications in clinical and home settings.**

Keywords— Breathomics, Respiratory Pattern Recognition, Image Processing, Machine Learning, Deep Learning.

I. INTRODUCTION

Exhaled human breath contains a rich array of volatile organic compounds (VOCs) that serve as indicators of physiological status and potential pathological conditions. The field of breathomics aims to harness this information for non-invasive diagnostics and continuous health monitoring. Traditional approaches to breath analysis, such as gas chromatography (GC), mass spectrometry (MS), and Fourier transform infrared spectroscopy (FTIR), are highly sensitive but suffer from limitations including high cost, operational complexity, and lack of portability—factors that restrict their use in real-time or remote health monitoring scenarios. Recent research has introduced visual breath analysis as an alternative, wherein breath condensation patterns formed on cold surfaces are captured and analyzed. These images often reveal structural and intensity variations that correlate with the depth and force of exhalation, offering a novel pathway for respiratory pattern classification. Unlike chemical sensors, this method is non-contact, cost-effective, and potentially deployable in everyday environments. In this work, we investigate a comprehensive framework for classifying breath patterns into three categories—normal, shallow, and deep—using both traditional machine learning models and modern deep learning architectures. While ML techniques rely on engineered statistical features extracted from the images, DL models learn hierarchical representations directly from raw pixel data, which could enhance classification performance and generalization. The models are evaluated under various conditions, including increased data complexity and real-world noise, to test their robustness and adaptability.

II. LITERATURE REVIEW

Andrés Arcentales, Beatriz F. Giraldo, Pere Caminal, Iván Diaz and Salvador Benito focused on analysing heart rate (RR intervals) and respiratory flow signals to predict outcomes of ventilator weaning in ICU patients. Using spectral methods like Power Spectral Density (PSD), Cross Power Spectral Density (CPSD), coherence analysis across different frequency bands, they found that cross-spectral features between heart and breathing signals, especially in very low frequency (VLF) and high frequency (HF), provided better group separation than individual signals. Their approach achieved 76.9% classification accuracy, highlighting the importance of heart–respiratory coupling in assessing weaning readiness [1].

Perena Gouma, Krithika Kalyanasundaram, Xiao Yun, Milutin Stanac´evic´ and Lisheng Wang presented the development of portable breath analyzer using a metal oxide-based nanosensor specifically designed for detecting low concentrations of ammonia in human breath. The researchers used sol-gel synthesized thin films of a stabilized metal oxide phase, which showed high selectivity and sensitivity to ammonia at part-per-billion levels even in humid and realistic breath-like environments. A simple handheld prototype was built where changes in sensor resistance are converted into binary signals to quickly screen for elevated ammonia levels, which can be linked to conditions like renal disease. This work demonstrates the potential for creating inexpensive, fast and noninvasive diagnostic tools using nanomaterials for breath analysis [2].

Satoru Momos developed a highly sensitive gas sensor based on a CuBr thin film, a p-type semiconductor, for detecting trace levels of ammonia (NH_3) in human breath. The sensor demonstrated rapid response, high selectivity against interfering gases, and potential for detecting biomarkers like nonanal related to lung cancer. Its compatibility with low-cost CMOS fabrication enables integration into compact, portable systems. This technology offers a promising path toward real-time, noninvasive breath analysis for early disease diagnosis [3].

Abel Torres and colleagues investigated a noninvasive method to assess diaphragm performance in severe COPD patients using mechanomyographic (MMG) signals recorded via chest-mounted capacitive accelerometers. MMG responses were compared to inspiratory pressure (IP) during a graded inspiratory loading task. Findings indicated that MMG amplitude increased with respiratory effort, while the signal's maximum frequency (fmax) declined, reflecting muscle fatigue. This suggests MMG may be a useful tool for monitoring respiratory effort and fatigue in COPD without invasive procedures [4].

Francesco Vicario proposed a noninvasive approach to estimate respiratory resistance (R) and elastance (E) in spontaneously breathing patients on mechanical ventilation. Unlike traditional methods that require patient passivity or esophageal pressure measurements, this technique uses standard airway pressure and flow data with a constrained optimization algorithm based on a simplified lung model. Incorporating physiological constraints on muscle pressure dynamics ensures robustness during active breathing. The method enables accurate, real-time monitoring of respiratory mechanics, potentially improving ventilator management without disrupting patient effort [5].

T. Lomonaco with his colleagues developed a portable computerized breath sampling system that records respiratory parameters such as mouth pressure, tidal volume, airflow, rate, and end-tidal CO_2 (pCO_2). It was used to examine how varying breathing rates (10, 30, 50 bpm) influence levels of VOCs like acetone, isopropyl alcohol, isoprene, and ethanol in exhaled breath. Findings showed that higher breathing rates reduce certain VOC concentrations, notably isoprene, due to decreased alveolar contribution. To correct for variability, pCO_2-based mathematical adjustments were applied. This system supports reliable breath analysis, particularly for patients with respiratory conditions [6].

M. Li developed a micro reactor for highly selective capture and analysis of ultra-trace ketones and aldehydes in human breath—key biomarkers for lung diseases. The device features micro pillars coated with a quaternary ammonium aminooxy salt that reacts with carbonyls via oximation, enabling over 99% capture efficiency. Unlike traditional adsorption methods, the compounds are recovered using solvent elution and analyzed by FT-ICR-MS, ensuring precise, non-destructive detection [7].

A. Existing System

Historically, breath analysis technologies have been dominated by chemical sensing methods that detect VOCs in exhaled air using advanced instrumentation. These systems, though highly accurate, require laboratory infrastructure, making them impractical for deployment in low-resource or home-based healthcare settings. Moreover, breath pattern recognition in current systems is either manual or based on thermal imaging, which requires costly infrared cameras and specialized processing algorithms. Emerging studies have considered the potential of visual condensation patterns as a low-cost, image-based signal for assessing respiratory activity. However, most existing implementations are rudimentary, often relying on subjective human observation without quantitative image analysis or automated classification pipelines. These limitations hinder scalability and clinical integration. Furthermore, traditional systems lack the capability to adapt to complex or noisy real-world conditions. For instance, environmental variations, subject motion, or inconsistencies in condensation surfaces can degrade performance. To address these challenges, our proposed system leverages artificial intelligence (AI)—including ML and DL models—to automatically extract and classify condensation pattern features, offering improved performance and reduced human dependency.

III. METHODOLOGY

The methodology for this study involves a multi-stage pipeline that integrates data preprocessing, feature extraction, model training, and performance evaluation. Two parallel approaches were developed: one using traditional machine learning (ML) algorithms with handcrafted features, and the other using deep learning (DL) architectures that operate directly on image data.

A. Data Description

This study uses a custom dataset of grayscale images capturing breath condensation patterns on cold surfaces (e.g., glass/mirror) under controlled conditions. Samples are manually labeled into three classes: Normal (D0), Shallow (D1), and Deep (D2) breathing, based on condensation density and spread. The dataset is organized into class-specific subfolders and split into a training set (90%) and a held-out test set (10%) to evaluate generalization. All images were resized to 128×128 pixels to ensure uniformity and compatibility with deep learning architectures. Class balance was maintained to prevent bias, enabling robust model training and evaluation. This low-resolution, image-based dataset supports development of scalable, non-contact respiratory monitoring solutions.

B. Proposed System

To enable traditional ML algorithms and to process the image data, statistical features were extracted from each image: i) **Mean Intensity:** Captures overall brightness related to breath density. ii) **Standard Deviation:** Measures texture variation. iii) **White Pixel Count:** Derived from thresholding the image, representing breath area coverage. All images were resized to a fixed dimension (128×128 pixels) and converted to grayscale for uniform processing. The extracted features were normalized and formatted into a tabular structure to train classifiers.

Machine Learning Models

The following traditional classifiers were implemented using Scikit-learn:

- Support Vector Machine (SVM)
- Random Forest
- Logistic Regression
- Decision Tree
- K-Nearest Neighbors (KNN)

Each model was trained using the extracted features from the training set and evaluated on the test set using accuracy, precision, recall, and F1-score.

Deep Learning Models

A parallel image-based pipeline was developed using deep learning. Raw images were rescaled and passed directly into several Convolutional Neural Network (CNN) architectures:

- Custom CNN
- MobileNetV2
- ResNet50
- AlexNet
- VGG16
- EfficientNetB0
- DenseNet121
- InceptionV3

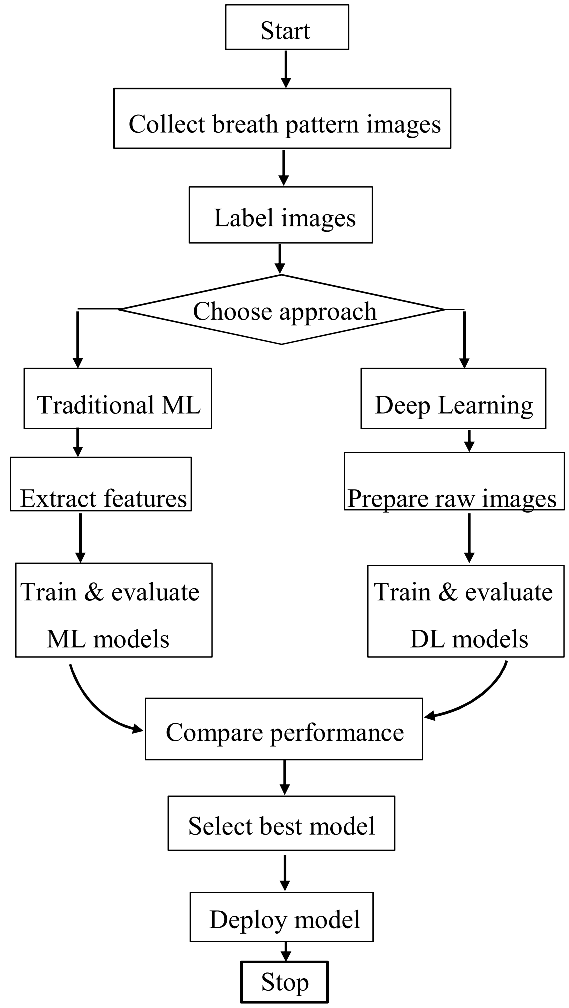

Fig. 1. Work Flow of the model.

The transfer learning models were initialized with ImageNet weights and fine-tuned using the breath dataset. Each model's top classification layer was replaced with a dense softmax layer corresponding to the three breathing classes. Categorical cross-entropy was used as the loss function, and the Adam optimizer was applied during training.

IV. RESULTS

To evaluate the effectiveness of the proposed machine learning (ML) and deep learning (DL) approaches for breath pattern classification, extensive experimentation was conducted using three different testing phases: Round-1 (10% base dataset), Round-2 (inbox dataset), and Round-3 (outbox dataset split into 25%, 50%, and 75% complexity levels). The models were assessed based on classification accuracy and detailed performance metrics such as precision, recall, and F1-score for each breath type: normal (D0), shallow (D1), and deep (D2).

A. Machine Learning Model Performance

Table I presents the accuracy of the traditional machine learning models. Random Forest achieved the best results across all test rounds. Support Vector

Machine, Logistic Regression, and K-Nearest Neighbors showed reduced performance as the complexity of the test data increased. The Decision Tree model produced high accuracy in Round-2 but was less in Round-3.

TABLE I: Accuracy of Machine Learning Models

Models	Round-1	Round-2	Round-3(25%)	Round-3(50%)	Round-3(75%)
SVM	90.31	36.36	78.08	78.08	65.42
Random Forest	92.35	82.52	83.56	95.4	72.4
Logistic Regression	89.31	34.97	78.49	72.9	58.9
Decision Tree	84.62	83.22	90.49	92.84	79.1
KNN	92.31	51.05	72.67	75.24	69.84

B. Deep Learning Model Performance

Table II shows the results of deep learning models. Deep learning models exhibited significantly better performance than traditional ML models. As shown in Table II, the Custom CNN and MobileNetV2 models demonstrated the most consistent and highest performance across all rounds. The Custom CNN achieved high accuracy and MobileNetV2 maintained high performance with accuracies of 93.53%, 94.8%, 94.2%, 96.72%, and 96.18%. InceptionV3 delivered the best peak performance in some rounds, with up to 100% accuracy in Round-1, though it showed variable results in later stages depending on data complexity. In contrast, models like EfficientNetB0 and ResNet50 underperformed compared to the lighter architectures, indicating that deeper networks may not generalize well under the constraints of limited dataset size.

TABLE II: Accuracy of Deep Learning Models

Models	Round-1	Round-2	Round-3(25%)	Round-3(50%)	Round-3(75%)
Custom CNN	95.25	96.23	94.14	95.43	97.22
MobileNetV2	93.53	94.8	94.2	96.72	96.18
ResNet50	82.77	90.05	84.83	88.97	84.46
EfficientNetB0	70.22	67.86	67.86	69.10	67.25
AlexNet	74.86	89.67	84.53	87.3	81.42
VGG16	78.9	86.38	89.25	75.28	73.29

C. Detailed Classification Reports for InceptionV3

Tables III to XI provide class-wise precision, recall, and F1-score for the InceptionV3 model. Across all rounds:

i. Normal breathing (D0) consistently showed high precision and recall, reaching 0.92 precision and 0.81 recall in the 50% test of Round-3.

ii. Shallow breathing (D1) had the lowest scores in earlier stages but improved over time, with 0.66 precision and 0.69 recall in the final test.

iii. Deep breathing (D2) showed increasing F1-scores in the final rounds, indicating better recognition in more complex scenarios. Overall accuracy improved from 62% in early tests to 77% in later, more challenging stages, demonstrating that the model adapts and scales well.

(25%): Inception V3

TABLE III: Classification Report (Test1) – 25%

	Precision	Recall	F1-score	Support
D0	0.83	0.69	0.75	58
D1	0.59	0.35	0.44	46
D2	0.51	0.79	0.62	53
Accuracy		0.62	157	
Macro Avg	0.65	0.61	0.61	157
Weighted Avg	0.65	0.62	0.62	157

TABLE IV: Classification Report (Test2) – 25%

	Precision	Recall	F1-score	Support
D0	0.85	0.62	0.72	174
D1	0.48	0.38	0.43	180
D2	0.47	0.69	0.56	180
Accuracy		0.57	534	
Macro Avg	0.60	0.57	0.57	534
Weighted Avg	0.60	0.57	0.57	534

TABLE V: Classification Report (Test3) – 25%

	Precision	Recall	F1-score	Support
D0	0.85	0.61	0.71	252
D1	0.51	0.28	0.36	144
D2	0.56	0.87	0.68	252
Accuracy		0.64	648	
Macro Avg	0.64	0.59	0.58	648
Weighted Avg	0.66	0.64	0.62	648

(50%): Inception V3

TABLE VI: Classification Report (Test1) – 50%

	Precision	Recall	F1-score	Support
D0	0.88	0.74	0.80	58
D1	0.57	0.54	0.56	46
D2	0.58	0.70	0.63	53
Accuracy		0.67	157	
Macro Avg	0.67	0.66	0.66	157
Weighted Avg	0.69	0.67	0.67	157

TABLE VII: Classification Report (Test2) – 50%

	Precision	Recall	F1-score	Support
D0	0.89	0.67	0.76	174
D1	0.47	0.54	0.50	180
D2	0.53	0.58	0.55	180
Accuracy		0.59	534	
Macro Avg	0.63	0.59	0.61	534
Weighted Avg	0.63	0.59	0.60	534

TABLE VIII: Classification Report (Test3) – 50%

	Precision	Recall	F1-score	Support
D0	0.92	0.81	0.86	252
D1	0.66	0.69	0.67	144
D2	0.74	0.81	0.77	252
Accuracy		0.78	648	
Macro Avg	0.77	0.77	0.77	648
Weighted Avg	0.79	0.78	0.79	648

(75%): Inception V3

TABLE IX: Classification Report (Test1) – 75%

	Precision	Recall	F1-score	Support
D0	0.87	0.83	0.85	58
D1	0.60	0.46	0.52	46
D2	0.57	0.72	0.63	53
Accuracy		0.68	157	
Macro Avg	0.68	0.67	0.67	157
Weighted Avg	0.69	0.68	0.68	157

TABLE X: Classification Report (Test2) – 75 %

	Precision	Recall	F1-score	Support
D0	0.85	0.67	0.75	174
D1	0.47	0.44	0.45	180
D2	0.48	0.61	0.54	180
Accuracy		0.57	534	
Macro Avg	0.60	0.67	0.58	534
Weighted Avg	0.60	0.57	0.58	534

TABLE XI: Classification Report (Test3) – 75%

	Precision	Recall	F1-score	Support
D0	0.88	0.83	0.85	252
D1	0.67	0.56	0.61	144
D2	0.73	0.84	0.78	252
Accuracy		0.77	648	
Macro Avg	0.76	0.74	0.75	648
Weighted Avg	0.77	0.77	0.77	648

D. Summary of Algorithm Accuracies

TABLE XII: Accuracy of Proposed Algorithms

Network	**Accuracy(%)**	**Level 1**	**Inbox**	**Outbox**
DenseNet121	Round1	83.33%	48.69%	66.05%
	Round2	36.36%	34.08%	41.36%
	Round3			
	25% Outbox	59.87%	49.25%	55.86%
	50% Outbox	58.60%	46.63%	63.12%
	75% Outbox	52.87%	45.88%	56.17%
InceptionV3	Round1	100.00%	47.9%	62.35%
	Round2	91.61%	75.66%	52.16%
	Round3			
	25% Outbox	62.42%	56.55%	63.73%
	50% Outbox	66.88%	59.36%	78.40%
	75% Outbox	68.15%	57.30%	77.31%

Table XII summarizes average accuracies across all rounds for DenseNet121 and InceptionV3. While DenseNet121 showed fluctuations and dropped to 34.08% at one point, InceptionV3 retained a more stable trajectory with accuracies peaking at 100% (Round-1) and staying above 60% even under challenging test conditions.

V. DISCUSSION

The experimental results highlight the clear advantage of deep learning methods over traditional ML techniques in classifying breath patterns. Among DL models, the Custom CNN and MobileNetV2 consistently outperformed others in all rounds, particularly in terms of stability and generalization. These models demonstrated high accuracy even on the more complex outbox dataset, showcasing their robustness to variations in input data. Deeper architectures like ResNet50 and EfficientNetB0 underperformed relative to lighter models. This may be attributed to the limited dataset size, which can lead to over fitting in models with a large number of parameters. In contrast, MobileNetV2 and InceptionV3 achieved high performance with relatively lower computational overhead, making them suitable for real-time applications on edge devices. In the ML domain, Random Forest emerged as the most effective classifier, offering stable results across multiple rounds, including the challenging outbox tests. Decision Tree showed competitive accuracy in specific scenarios (e.g., Round-2) but lacked consistent performance. Other models such as SVM and Logistic Regression experienced significant drops in accuracy under complex test conditions, indicating limitations in scalability. Overall, the findings support the use of deep learning—particularly lightweight transfer learning models—as a scalable and accurate solution for breath pattern classification using visual data.

VI. CONCLUSION

This study presented a comprehensive analysis of image-based breath pattern classification using both traditional machine learning and deep learning models. By leveraging breath condensation images as a proxy for respiratory activity, the study demonstrated the feasibility of non-contact and non-invasive respiratory monitoring using accessible imaging techniques. The comparative evaluation across three progressive testing rounds—ranging from held-out test sets to complex, unseen samples—revealed that deep learning models, especially Custom CNN and MobileNetV2, deliver consistently high performance. These models not only generalize well but also maintain accuracy under real-world conditions, validating their potential for clinical deployment. Among traditional models, Random Forest showed notable resilience, suggesting its usefulness in settings with limited computational resources. Future work will explore the integration of temporal features, real-time video streams, and thermal imaging to further enhance system accuracy. Expanding the dataset with more diverse subjects and environmental conditions will also improve model robustness. Ultimately, the proposed system aims to contribute to the development of affordable, accessible respiratory health monitoring solutions suitable for use in telemedicine, emergency triage, and home healthcare environments.

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